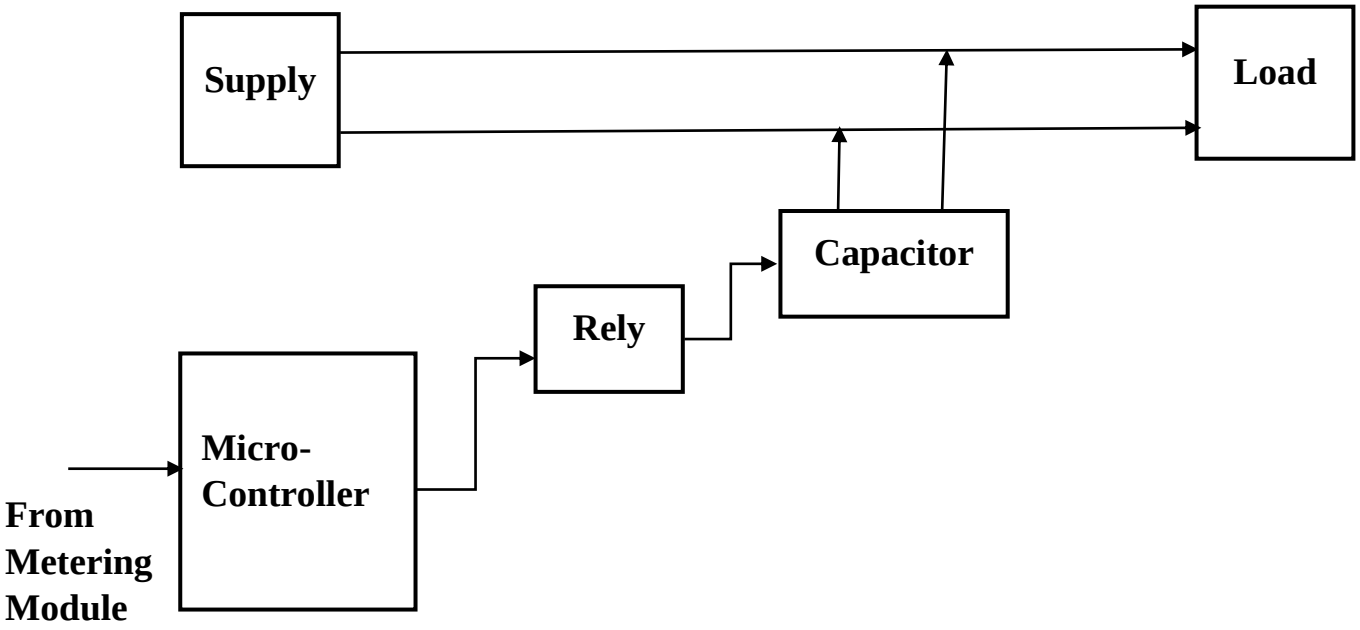
Automatic Power Factor Correction (APFC) System for Residential Purpose:



Block Diagram for APFC

An **Automatic Power Factor Correction (APFC)** system is used to improve the power factor of a residential or industrial electrical system. The main function of the APFC system is to reduce reactive power and ensure efficient use of electrical energy. It helps in avoiding penalties from electricity providers for poor power factor and reduces energy losses.

For a residential setup with a **5 – 7 kW maximum load**, you can implement a capacitor-based power factor correction. In this case, a single capacitor can be connected as a capacitor bank to adjust the power factor.

Understanding Power Factor:

- **Power Factor (PF):** It is the ratio of real power (kW) to apparent power (kVA). It is a measure of how efficiently electrical power is being used.
- A low power factor indicates poor efficiency, leading to higher electricity bills and potential penalties.
- The ideal power factor is **1.0**, meaning all power is being used effectively.

Objective of APFC:

- The goal is to bring the power factor as close to **1.0** as possible.
- Residential loads like motors, air conditioning units, pumps, and other inductive loads tend to lower the power factor due to their reactive power demand.

Single Capacitor Bank for Power Factor Correction:

A **capacitor bank** stores and releases electrical energy, offsetting the inductive load in the system and improving the power factor.

Capacitor Calculation for Power Factor Correction:

To calculate the required capacitance for a capacitor bank to correct the power factor, the following steps can be taken:

1. **Load Details:**
   - Maximum real power load Pload = **5 – 7 kW**
   - Assumed initial power factor (before correction) = **0.75**
   - Desired power factor (after correction) = **0.99**
2. **Power Factor Correction Formula:** The reactive power required to correct the power factor is calculated using the following formula:

   Q = Pload×(tan(cos^{−1}(PFinitial))−tan(cos^{−1}(PFdesired)))

   Where:

   - Pload = Real power in kW (5kW)
   - PFinitial = Initial power factor (0.75)
   - PFdesired = Desired power factor (0.99)
3. **Step-by-Step Calculation:**
   - **Step 1:** Find the angle corresponding to the initial power factor:

     θi=cos^{−1}(0.75) = 41.40

     The tangent of this angle:

     Tan(θi) = tan(41.40) = 0.88

   - **Step 2:** Find the angle corresponding to the desired power factor:

     θf=cos^{-1}(0.99) = 8.10

     The tangent of this angle:

     Tan(θf) = tan(8.10)=0.14

   - **Step 3:** Calculate the required reactive power Q

$$Q = 5kW \times (0.88 - 0.14)$$
$$= 3.7\ kVAR$$

4. **Capacitor Value Calculation:** To determine the required capacitance (CCC) in farads to provide this reactive power, we can use the formula:

   C = Q / (2π x F x V^{2})

   C = 3.7 / (2π x 50 x 220^2)
   C = 243.46 uF

Thus, a **capacitor bank of approximately 243.46 µF** is required to correct the power factor from 0.75 to 0.99 for a 5kW load in a residential system.

Key Components of an APFC System:

1. **Capacitor Bank:**
   - A single capacitor with a value of **243.46 µF** (or a combination of capacitors to achieve this value) can be used as a capacitor bank.
2. **APFC Controller:**
   - An **automatic power factor controller** monitors the power factor continuously and switches the capacitor in and out of the circuit to maintain the desired power factor.

Advantages of an APFC System:

- **Reduces electricity bills**: Improves power efficiency and avoids penalties for poor power factor.
- **Improves voltage regulation**: A better power factor reduces voltage drops in the system.
- **Increases equipment lifespan**: Reduces the strain on electrical appliances and motors.
- **Compact design**: Single capacitor-based systems are simpler and easier to implement in residential setups.

Conclusion:

For a residential setup with a maximum load of 5 – 7 kW, installing a **240 - 350 µF capacitor** and an APFC controller can significantly improve the power factor, leading to energy savings and better equipment efficiency. This setup is cost-effective and simple for home usage, enhancing both system performance and power quality.

Relay Specifications:

To control a 240-350 µF capacitor in an AC circuit with a voltage of 220 VAC, you'll need a relay with specific characteristics.

1. **Type**: Electromechanical relay (EMR) or solid-state relay (SSR).
    - **EMR**: Good for lower frequencies and provides isolation.
    - **SSR**: Better for higher switching speeds and longer life, but typically more expensive.
2. **Coil Voltage**: This should match your control circuit voltage (e.g., 220 VAC).
3. **Contact Rating**:
    - **Voltage Rating**: Must be rated for at least 250 VAC to handle 220 VAC safely.
    - **Current Rating**: Calculate the current using the formula:

      $I = V / Z$

      For 220 VAC and assuming 50 Hz, this needs to be considered for inrush current.
4. **Contact Configuration**: Typically, a Single Pole Double Throw (SPDT) or Double Pole Double Throw (DPDT) relay can be used.
5. **Inrush Current Rating**: Make sure the relay can handle inrush currents, which can be significantly higher than normal operating currents.

Example Specifications

- **Coil Voltage**: 220 VAC (or suitable low voltage for your control circuit)
- **Contact Rating**: 15A at 250 VAC (to accommodate inrush current)
- **Type**: SPDT or DPDT
- **Inrush Current**: Rated for higher than the expected inrush current of the capacitor.